How-To Library

Informatica

Frequently Asked Questions for PowerExchange for Kafka for PowerCenter

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Abstract

This article describes frequently asked questions about using PowerExchange for Kafka for PowerCenter to read data from and write data to Kafka topics.

Supported Versions

- PowerExchange for Kafka for PowerCenter 10.4.1

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General Questions

What is Apache Kafka?

Apache Kafka is an open-source scalable, fault tolerant, and distributed data streaming platform. You can process large amounts of data quickly by using Kafka. Kafka can serve as an interim staging area for data consumed by various types of downstream consumer applications.

Kafka runs as a cluster comprised of one or more servers each of which is called a broker. Kafka brokers stream data in the form of messages. These messages are published to a topic.

Kafka topics are divided into partitions. This gives better throughput and could be used to scale the number of messages processed. Message ordering is guaranteed only within partitions.

How can I configure a Kafka cluster?

To configure a Kafka cluster, you must have at least three nodes or Kafka brokers. You must create configuration files for each of the broker and start the nodes on ZooKeeper. For more information about configuring a Kafka cluster, see Kafka documentation.

Which is the latest available version of PowerExchange for Kafka for PowerCenter?

PowerExchange for Kafka 10.4.1 is the latest available version for PowerCenter.

How do I install PowerExchange for Kafka for PowerCenter?

PowerExchange for Kafka is packaged with the Informatica services 10.4.1 installer.

What does PowerExchange for Kafka use to access a Kafka cluster?

PowerExchange for Kafka uses the Kafka Consumer and Producer API version 0.11.0.0 to connect to a Kafka cluster.

What Kafka Broker version does PowerExchange for Kafka use?

PowerExchange for Kafka uses the Apache 0.10.1.1 and above as the Kafka messaging broker version.

What are the variants of Kafka that PowerExchange for Kafka support?

Yes. PowerExchange for Kafka for PowerCenter supports the following variants of Kafka clusters:

- Apache Kafka
- Confluent Kafka

Apache Kafka or Confluent Kafka clusters can run locally or on Cloud.

What information do I need to create a Kafka connection?

Before you create a Kafka connection, you must have the following information:

Kafka Broker List

The IP address and port combinations of the Kafka messaging system broker list. You can enter multiple comma-separated IP address and port combinations.

Retry Timeout in seconds

Number of seconds the Integration Service attempts to reconnect to the Kafka broker to write data.

Kafka Broker Version

The Kafka messaging broker version. PowerExchange for Kafka supports **Apache 0.10.1.1 and above** versions of the Kafka broker.

Additional Connection Properties

Comma-separated list of connection properties to connect to the Kafka broker. You can specify the configuration properties required to connect to a Kerberised Kafka cluster. You can also specify SSL specific properties to connect to a Kafka broker over SSL.

SSL Mode

Determines whether the PowerCenter Integration Service establishes a secure connection to the Kafka broker. You can select disabled, require, one-way, or two-way as the SSL mode.

SSL TrustStore File Path

The complete path and file name of the truststore file. The truststore file contains the SSL certificate that the Kafka cluster validates against the Kafka broker certificate.

SSL TrustStore Password

The password for the truststore file

SSL KeyStore File Path

The complete path and name of the Java keystore file. The keystore file contains the certificate that the Kafka broker validates against the Kafka cluster certificate.

SSL KeyStore Password

The password for the keystore file.

For more information about creating a Kafka connection, see the *PowerExchange for Kafka User Guide for PowerCenter*.

What are the operations that PowerCenter supports for PowerExchange for Kafka?

You can use PowerExchange for Kafka to read from and insert data to Kafka topics.

What are the file formats supported by PowerExchange for Kafka?

PowerExchange for Kafka supports the following file formats:

- Binary
- JSON
- Avro

When you import a Kafka topic, you can specify the format in which the PowerExchange for Kafka reads data from or writes data to a Kafka topic. You can specify JSON or Avro as format. When you specify JSON format, you must provide a sample JSON schema file. When you specify Avro format, provide a sample Avro schema as a `.avsc` file.

Does PowerExchange for Kafka supports commit interval?

No. Commit interval is not applicable when you configure a Kafka session.

What is the default file format when I import a Kafka topic?

When you import a Kafka topic and do not select a format type, PowerExchange for Kafka imports the data from the Kafka topic in binary format.

How can I configure Topic Pattern to read data from Kafka topics in a PowerCenter session?

When you configure a Kafka session to read data from a Kafka topic, you can specify a regular expression for the topic name pattern in the **Topic Pattern** source session property in the **Mapping** tab.

For more information on how to configure topic patterns, see the KB article 565201.

How do I access the sample data or schema of a Kafka topic?

You can preview the Kafka source or target data when you import a Kafka topic.

How can I configure a secured connection to Kafka?

PowerExchange for Kafka supports one-way SSL, two-way SSL, and Kerberos Authentication protocols. You can configure SSL specific properties in the Kafka connection to connect to a Kafka broker over SSL. You can also specify the configuration properties required to connect to a Kerberised Kafka cluster in the Additional Connection Properties in the Kafka connection. You can also specify the configuration properties required to connect to a Kerberised Kafka cluster in the Consumer Configuration Properties or Producer Configuration Properties in the Kafka session.

How can I configure properties specific to a Kafka consumer or producer?

You can use the following session properties to configure properties specific to a Kafka consumer or producer:

- Consumer Configuration Properties. Specify configuration properties specific to a Kafka consumer. For more information about consumer configuration properties, see Kafka documentation.
- Producer Configuration Properties. Specify configuration properties specific to a Kafka producer. For more information about producer configuration properties, see Kafka documentation.

How can I configure Consumer Configuration Properties?

You can configure the Consumer Configuration Properties for a Kafka source on the **Mapping** tab. If you read data from a Kafka cluster that is configured for Kerberos authentication, specify the following property to override the **Additional Connection Properties** specified in the Kafka connection:

```
security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=GSSAPI
```

How can I configure Producer Configuration Properties?

You can configure the Producer Configuration Properties for a Kafka target on the **Mapping** tab. If you write data to a Kafka cluster that is configured for Kerberos authentication, include the following property to override the **Additional Connection Properties** specified in the Kafka connection:

```
security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=GSSAPI
```

You can provide the **Batch Flush Time in milliseconds** and **Batch Flush Size in bytes** properties in the producer configuration properties.

For example,

```
batch.size=100,linger.ms=5
```

What are the message recovery strategies supported for a Kafka session?

When you configure a session, you can configure the **Recovery Strategy** property on the **Properties** page of the Session task.

You can select one of the following recovery strategies:

- **Resume from the last checkpoint.** The PowerCenter Integration Service saves the session state of operation and maintains target recovery tables. If the session aborts, stops, or terminates, the PowerCenter Integration Service uses the saved recovery information to resume the session from the point of interruption. You cannot configure a session with a resume strategy if it uses mapping variables.
- **Restart task.** The PowerCenter Integration Service runs the session again when it recovers the workflow. When you recover with restart task, you might need to remove the partially loaded data in the target or design a mapping to skip the duplicate rows.
- **Fail task and continue workflow.** When the PowerCenter Integration Service recovers a workflow, it does not recover the session. The session status becomes failed, and the PowerCenter Integration Service continues running the workflow.

How can I terminate a Kafka session?

You can set the following JVM option as a custom property for the PowerCenter Integration Service to abort a session:

```
-DINFA_KAFKA_PC_SUCCESS_ROW_COUNT = <number of success rows>
```

The value of the property determines the number of success rows after which the PowerCenter Integration Service stops reading messages from a Kafka source or writing data into a Kafka target and ends the session.

What are the data sources and targets that can be supported by PowerExchange for Kafka?

PowerExchange for Kafka supports data sources and targets supported by PowerCenter.

Does PowerExchange for Kafka support File Based Recovery Mechanism?

Currently, File Based Recovery for PowerExchange for Kafka is available for technical preview in 10.4.0 release. PowerExchange for Kafka will support File Based Recovery in a future release.

Which Kafka data types does PowerExchange for Kafka support?

For information about the various data types that PowerExchange for Kafka supports, see the chapter "Kafka Data Type Reference" in the *Informatica PowerExchange for Kafka 10.4.1 User Guide for PowerCenter*.

Author

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